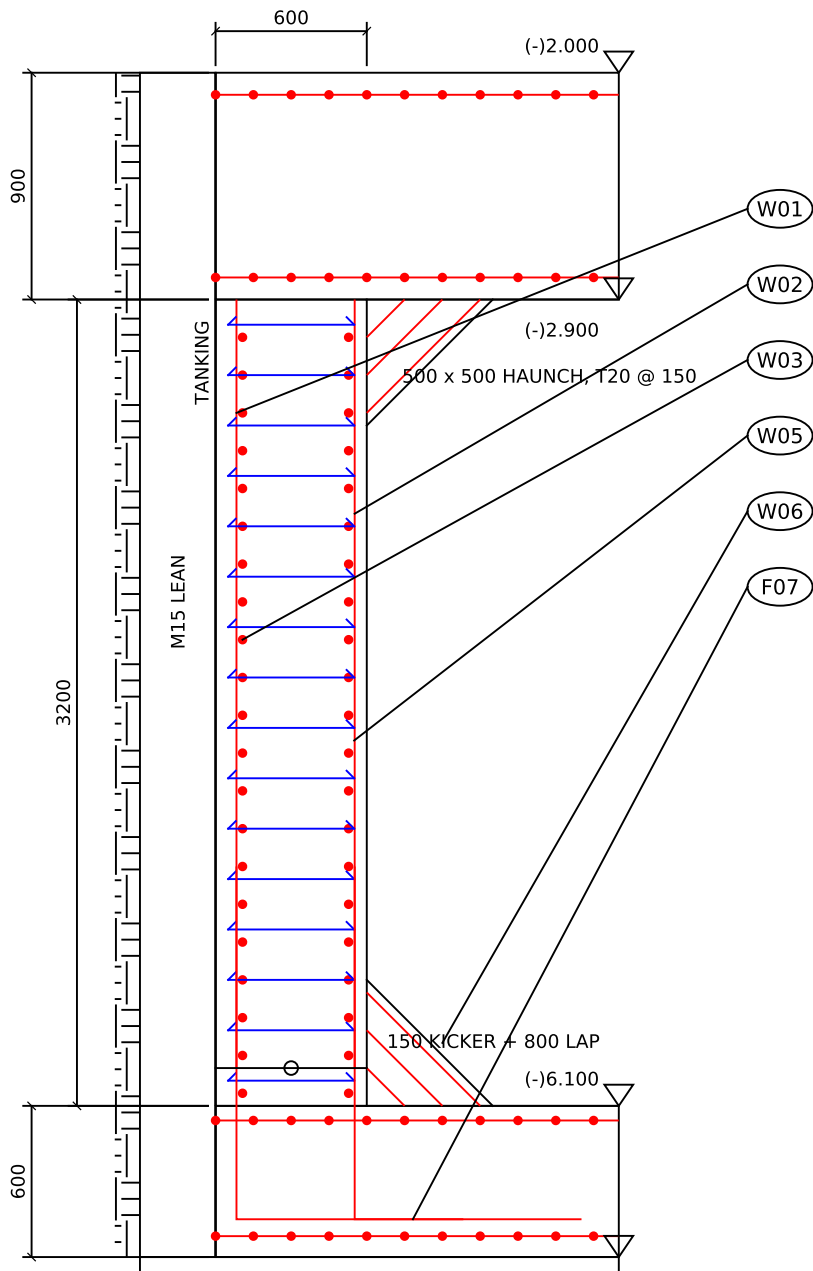
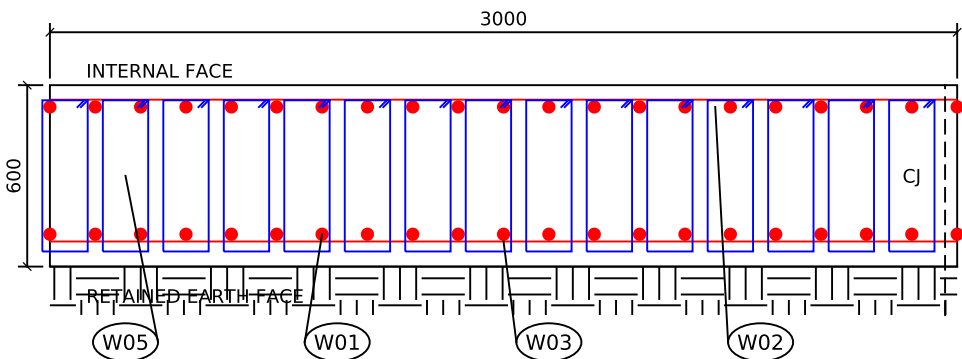




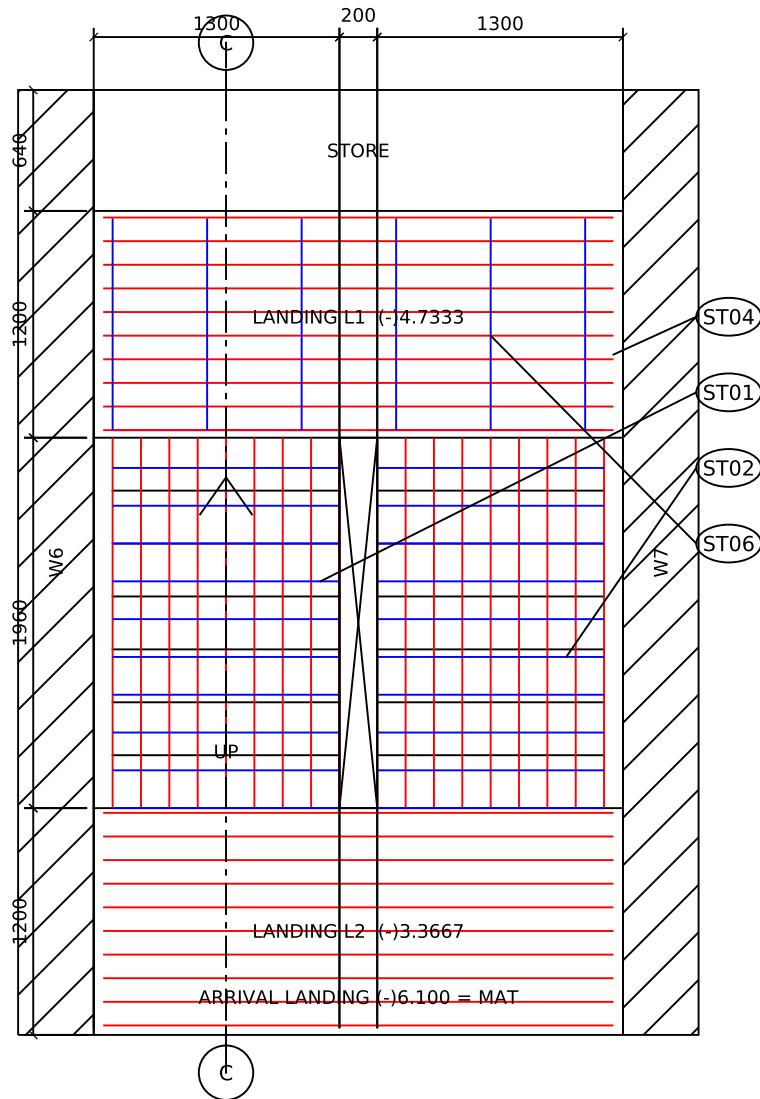
1 600 SHEAR WALL - VERTICAL SECTION
1 : 30 W1 / W2 / W3 / W4. T16 @ 150 EF EW, COVER 75 / 40



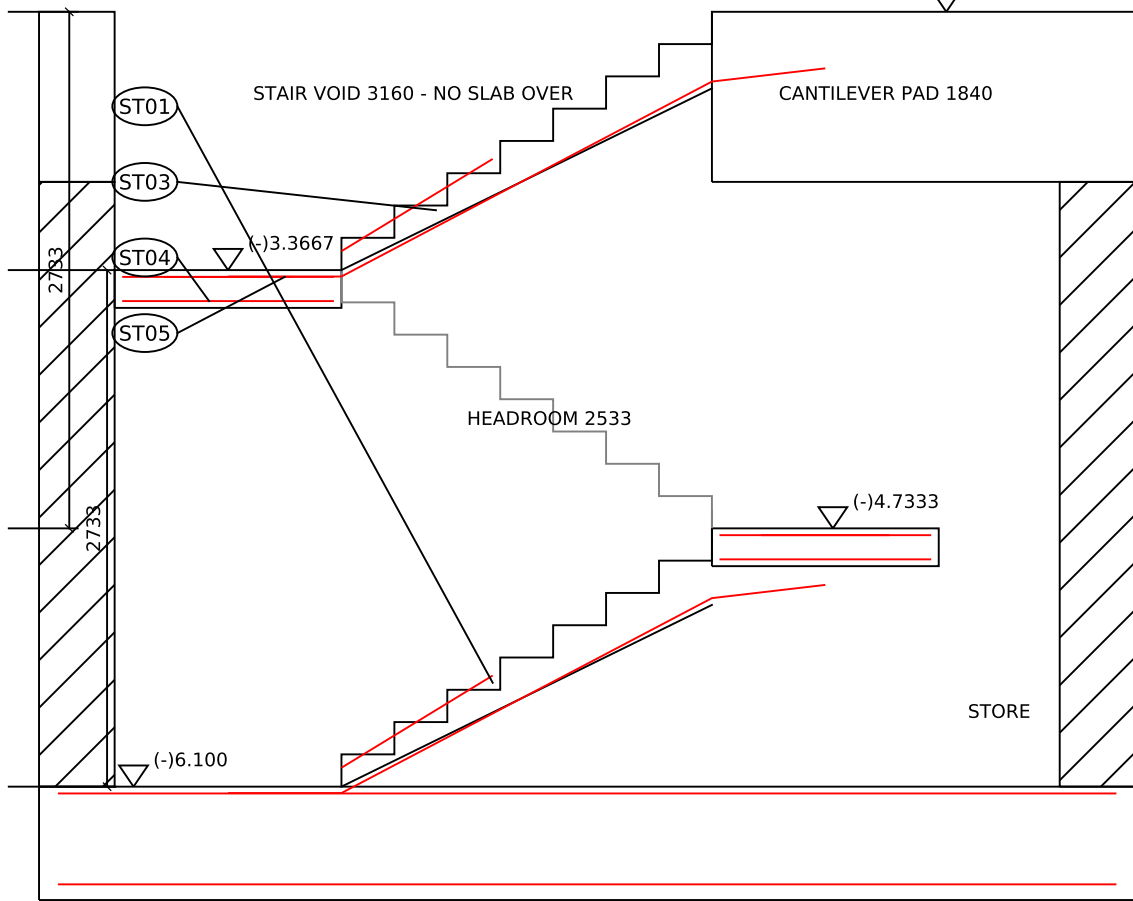
2 600 SHEAR WALL - PART PLAN
1 : 25 HORIZONTAL SECTION AT (-)4.500

WALL AND STAIR - CONSTRUCTION REQUIREMENTS

- WALL STARTERS F07: T16 @ 150 EACH FACE WITH A 900 mm HORIZONTAL LEG INTO THE MAT, lapped 800 (50 phi) above a 150 kicker. The starter, not the wall bar, is what makes the mat / wall joint work.
- HAUNCHES 500 x 500 AT EVERY WALL / ROOF AND WALL / MAT JUNCTION, with diagonal T20 @ 150 (mark W06).
- TWO WATERSTOPS AND A WELDED Cu / GALVANISED EMP STRAP AT EVERY CONSTRUCTION JOINT (approximately 6 m centres). Reinforcement is FULLY CONTINUOUS through the joint - it is not a movement joint.
- COVER: 75 mm to a face cast against the blinding or trimmed rock; 50 mm to a formed earth face; 40 mm internally; 30 mm in the stair shaft. The 5 mm reduction IS 456 Table 16 permits for M35 and above is DELIBERATELY NOT TAKEN.
- MAX BAR SPACING 150 IN BOTH CURTAINS - AN EMP REQUIREMENT, STRICTER THAN IS 456 Cl. 26.3.3. No bar may be displaced to clear a service; any clash is an RFI.
- THE MAIN STAIRCASE IS DESIGNED TO IS 456 WITH NORMAL PARTIAL FACTORS. It is inside the protective envelope and is NOT a blast element.
- STAIR FINISHES AND THE 1100 GUARDING TO THE VOID FREE EDGE (NBC 2016 Pt 4) ARE NOT SHOWN ON THIS SHEET - see the architectural package.



3 MAIN STAIRCASE - REINFORCEMENT PLAN
1 : 40 BAY 7. 24R @ 170.8333 / 280, 3 FLIGHTS x 8



4 MAIN STAIRCASE - SECTION C-C
1 : 40 CUT ON FLIGHT A AT X 15900, LOOKING EAST

MAIN STAIRCASE - THE GEOMETRY IS FROZEN

24 RISERS AT 170.8333 | TREAD 280 | 3 FLIGHTS x 8 RISERS | TOTAL RISE 4100
FLIGHT WIDTH 1200 | WELL 200 | WAIST 200 | HEADROOM 2533
LANDING L1 (-)4.7333 L2 (-)3.3667 ARRIVAL (-)6.100 = THE MAT SURFACE
TOTAL RISE CLOSES EXACTLY: (-)6.100 TO (-)2.000 = 4100 = 24 x 170.8333
NO RISER, TREAD, LANDING LEVEL, FLIGHT WIDTH, WELL OR HEADROOM ON THIS SHEET
MAY BE ALTERED. ANY CHANGE IS A NEW INSTRUCTION, NOT A DETAILING DECISION.

WALL SCHEDULE

WALL	THK	LENGTH	COVER	d	MAIN BARS	LINKS
W1 SOUTH PERIMETER	600	22000	75/50/40	517	T16 @ 150 EF EW	T12 CLOSED @ 200
W2 NORTH PERIMETER	600	22000	75/50/40	517	T16 @ 150 EF EW	T12 CLOSED @ 200
W3 WEST END	600	6200	75/50/40	517	T16 @ 150 EF EW	T12 CLOSED @ 200
W4 EAST END	600	6200	75/50/40	517	T16 @ 150 EF EW	T12 CLOSED @ 200
ALL FOUR 600 WALLS: xu/d = 0.089, UTILISATION 68 %, BLAST GOVERNS 4.6 : 1						
OTHER WALLS - SCHEDULED HERE, DETAILED ELSEWHERE, NOT DRAWN ON THIS SHEET						
W5 BAY 5 / 6	200	5000	40	154	T12 @ 150 EF EW	- NOMINAL
W6 BAY 6 / 7 (M1)	400	5000	50/40	342	T20 @ 150 EF EW	T12 4L @ 200
W7 BAY 7 / 8 (M1)	400	5000	50/40	342	T20 @ 150 EF EW	T12 4L @ 200
W8 PARTITIONS x 4	110	5000	25	-	A252 MESH B/F	- NON-STRUCTURAL

BAR MARK SCHEDULE - 600 PERIMETER SHEAR WALLS

MARK	BAR	SPACING	LOCATION	CUT mm	No.
W01	T16	150	VERTICAL, EARTH FACE	3875	360
W02	T16	150	VERTICAL, INTERNAL FACE	3875	360
W03	T16	150	HORIZONTAL, EACH FACE	11350	176
W04	T16	150	HORIZONTAL, EACH FACE	7380	88
W05	T12	200 x 200	CLOSED LINKS, 200 x 200 GRID	1540	4320
W06	T20	150	HAUNCH DIAGONALS, WALL/ROOF + WALL/MAT	2307	720

MAIN STAIRCASE - ELEMENT SCHEDULE

ELEMENT	THK	d	MAIN	DISTRIB.	TOP STEEL
FLIGHTS 1, 2, 3 (WAIST)	200	164	T12 @ 150 (754)	T10 @ 200	T12 @ 150, 800 IN
LANDING L1 (-)4.7333	200	164	T12 @ 125 BTM (905)	T10 @ 200	T12 @ 125, 900 EACH END
LANDING L2 (-)3.3667	200	164	T12 @ 125 BTM (905)	T10 @ 200	T12 @ 125, 900 EACH END
ARRIVAL LANDING (-)6.100	-	-	= THE MAT SURFACE	-	NO SEPARATE SLAB

BAR MARK SCHEDULE - MAIN STAIRCASE

MARK	BAR	SPACING	LOCATION	CUT mm	No.
ST01	T12	150	MAIN BARS (x3 FLIGHTS)	3496	24
ST02	T10	200	DISTRIBUTION (x3 FLIGHTS)	1140	36
ST03	T12	150	TOP STEEL, 800 INTO THE FLIGHT	1280	48
ST04	T12	125	BOTTOM MAIN	3200	20
ST05	T12	125	TOP, 900 FROM EACH SUPPORT	1380	40
ST06	T10	200	DISTRIBUTION	1140	30

MAIN STAIRCASE - NBC 2016 PART 4 GEOMETRY CHECK

ITEM	PROVIDED	LIMIT	VERDICT
RISER	170.8333 mm	NOT MORE THAN 190	PASS
TREAD / GOING	280 mm	NOT LESS THAN 250	PASS
FLIGHT WIDTH	1200 mm	NOT LESS THAN 1000	PASS - STRETCHER CAPABLE
RISERS PER FLIGHT	8	12 PREFERRED MAX	PASS
HEADROOM	2533 mm	NOT LESS THAN 2200	PASS
TOTAL RISE	4100 = 24 x 170.8333	(-)6.100 TO (-)2.000	CLOSES EXACTLY

DESIGN BASIS - THIS SHEET

600 PERIMETER SHEAR WALLS W1 - W4 d = 600 - 75 - 8 = 517
Blast 383 kPa (Ka 1.0) against static earth + water 83.2 kPa at the base -> BLAST GOVERNS 4.6 : 1
Mp = w Ln2 / 16 = 383 x 3.2002 / 16 = 245.1 kN/m
Mu,lim = 0.133 fck.dyn b d2 = 1556 kN/m OK
Ast,req (fy.dyn 625, fck.dyn 43.75) = 894 mm2/m
MINIMUM STEEL, ALL THREE CHECKED:
IS 456 Cl. 32.5(a) vertical 0.0012 x 600 = 720 mm2/m
IS 456 Cl. 32.5(b) horizontal 0.0020 x 600 = 1200 mm2/m
IS 3370 Pt 2 surface zone 0.35 % x 250/face = 875 mm2/m/face
PROVIDED T16 @ 150 EF EW = 1340 mm2/m
Mu 362.8 kN/m, UTILISATION 68 %, xu/d = 0.089 < 0.46
SHEAR V at d = 383 (1.600 - 0.517) = 414.8 kN/m
tv 0.802 | tc 0.375 (pt 0.259 %) | tc,max 3.70 OK
Vus 220.8 kN/m ; Asvsv 0.982 ; Cl. 26.5.1.5 limit 300
PROVIDED T12 CLOSED LINKS @ 200 (Asvsv 1.131)
AXIAL N = 1168 kN/m ; f1 95 N/mm2 = 11 % of 0.4 fck.dyn
-> P-M INTERACTION NOT CRITICAL
CRACK IS 3370 Pt 2 limit 0.2 mm ; 875 required < 1340 provided
SEISMIC Ah 0.075 ; Vb 1050 kN ; 525 kN per long wall
tau = 0.063 N/mm2, NEGLIGIBLE
IS 13920 Cl. 10.4 BOUNDARY ELEMENTS ARE NOT TRIGGERED

WHY 600 - STATED HONESTLY. The wall is NOT strength-governed at 68 %. 600 is set by (i) the 75 mm cover, (ii) congestion - two curtains plus closed links plus waterstops plus cast-in frames, (iii) IS 3370 crack control under sustained hydrostatic load, (iv) the EMP double curtain at 150, (v) the 1168 kN/m axial path.

MAIN STAIRCASE, BAY 7 INSIDE the protective envelope, and NOT a blast element. The shaft pressurises, but that pressure acts on the shaft WALLS and on the blast DOORS - not as a net load on a slab that is open on both faces. Designed to IS 456 with normal partial factors, NOT with IS 4991 dynamic strengths.

FLIGHT, WAIST 200 theta = arctan(170.833 / 280) = 31.4 deg
waist self 5.858 + steps 2.135 (Cl. 33.2) + finishes 1.000
+ live 5.000 = 13.99 kPa -> wu = 21.0 kPa
Leff = going 1960 + 600 + 600 = 3160 (IS 456 Cl. 33.1(b))
d = 200 - 30 - 6 = 164 ; M = wu Leff2 / 8 = 26.2 kN/m
Ast,req 380 ; Cl. 26.5.2.1 min 0.12 % x 200 = 240
PROVIDED T12 @ 150 (754) -> Mu 50.3 kN/m, UTILISATION 52 %
T10 @ 200 DISTRIBUTION ; T12 @ 150 TOP, 800 in, Ld 480
DEFLECTION Leff/d 19.3 ; fs 146 ; pt 0.46 % ; MF 1.35 -> 27
-> permissible 38 > 19.3
SHEAR V 33.2 ; tv 0.202 < tc 0.479 x k 1.20 = 0.575 NO LINKS

LANDINGS L1 / L2, 200 THK, SPANNING ACROSS THE SHAFT
self 7.50 + finishes 1.50 + live 7.50 + flight reaction 27.70
= 44.20 kPa ; Left = 2800 clear + 200 bearing = 3000 (Cl. 22.2(a))
M = 49.7 kN/m ; Ast,req 745
T12 @ 150 would give 754 = 100 % - TOO TIGHT.
PROVIDED T12 @ 125 BOTTOM (905) -> Mu 59.5, UTILISATION 84 %
DEFLECTION 3000 / 164 = 18.3 ; fs 239 ; pt 0.552 % ; MF 1.35 -> 27
SHEAR V 66.3 kN/m ; tv 0.404 < tc 0.519 x 1.20 = 0.622

THE LANDINGS PROP W6 AND W7 AT (-)4.733 AND (-)3.367, BUT ONLY OVER THEIR 1200 DEPTH. The 400 wall design does NOT rely on that prop.

UNDERGROUND CBRN-HARDENED
BLAST-RESISTANT PROTECTIVE
STRUCTURE + SENTRY POST
PUNE, MAHARASHTRA
STRUCTURAL - PHASE 2 REV A + M1

NOTES

- ALL DIMENSIONS ARE IN MM. ALL LEVELS ARE IN METRES RELATIVE TO FINISHED SITE GRADE 0.000, NEGATIVE DOWNWARDS.
- WRITTEN DIM TO BE FOLLOWED ONLY. DRG NOT TO BE SCALED.
- THIS DRG IS TO BE READ IN CONJUNCTION WITH ALL RELEVANT ARCH AND STR DRGS.
- CONCRETE M35, w/c NOT MORE THAN 0.45. REINFORCEMENT Fe500D TO IS 1786:2008. Ld 40 PHI, LAPS 50 PHI, ALL STAGGERED SO THAT NOT MORE THAN 50 % ARE SPLICED AT ONE SECTION.
- THE 600 WALL IS NOT STRENGTH-GOVERNED (68 %), 600 IS SET BY COVER, CONGESTION, IS 3370 CRACK CONTROL, THE EMP DOUBLE CURTAIN AT 150 AND THE 1168 kN/m AXIAL PATH.
- THE MAIN STAIRCASE IS INSIDE THE PROTECTIVE ENVELOPE AND IS NOT A BLAST ELEMENT. IT IS DESIGNED TO IS 456 WITH NORMAL PARTIAL FACTORS, NOT WITH IS 4991 DYNAMIC STRENGTHS.
- IS 13920:2016 HAS BEEN CHECKED FOR THE BOX: Cl. 10.4 BOUNDARY ELEMENTS ARE NOT TRIGGERED (WALL SHEAR 0.063 N/mm2) AND NO DUCTILE-DETAILING CLAUSE GOVERNS ANY BAR ON THIS SHEET. CONTRACTOR TO CHECK AND VERIFY THE DRG BEFORE EXECUTION.
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STRUCTURAL REINFORCEMENT DETAILING - 600 SHEAR WALL & MAIN STAIRCASE

CBRN HARDENED UG OPS ROOM

SHEET 07

Project number 01

Date 16 SEP 2026

Drawn by SYN 01

Checked by DR IR CHAUDHARI

STR007

Scale As indicated